

Phonebook Lookup — Group Report (Parts A, B, D)

Group G4 — Searching & Hash tables (§2.2) · Linear · Binary · Interpolation · Hashing & collisions · Demo: phone-book lookup ($n \geq 100k$) · Members: · Build/run & machine spec: README.md

Parts A, B, D of C0B_Group_Project_Guide.pdf. Contribution & AI-use declarations in Appendix.

Part A — Position & purpose

A.1 Syllabus map

Structure	Ref	LO	Built on	Built on top
Linear search	§2.2.2 p.3	LO2	unsorted array	— (baseline; inner loop of hash chains)
Binary search	§2.2.3 p.3	LO2	sorted array	ancestor of BST/B-trees
Interpolation search	§2.2.4 p.4	LO2	sorted array, uniform keys	—
Hashing & collisions	§2.2.5 p.4	LO2+LO3	array buckets + linked-list chaining	dicts, caches, DB hash indexes

Refs: course syllabus 3-Data Structure and Algorithm-CLC.pdf (INT1306_CLC, 2026-09-11); CLRS 4th ed. Ch.11 Hash Tables (§11.1–11.4), MIT Press 2022 (3rd ed. pp.253–280).

A.2 Motivation

Linear search — finds a record with no order/index by scanning until `phone==target` (phonebook.cpp:250). Fine for dozens, unusable at 100k (up to 100k compares per lookup). Baseline every other structure improves on; survives as hash-chain scan (hashtable.cpp:hashSearch). Wrong when lookups are frequent/large or data is sorted (binary dominates).

Binary search — exploits order: each probe halves the range, so 100k needs ≤ 17 compares vs 100k. Wastes sorted order if you scan linearly. Built on sorted array (buildSortedIndex() $O(n \log n)$ once; lowerBound $O(\log n)$ + vector::insert shift $O(n)$ per insert, phonebook.cpp:177). Ancestor of tree indexes. Wrong when unsorted (sorting cost > scan), frequently mutated ($O(n)$ insert to keep order vs hash $O(1)$), or keys have no order (exact phone match → hash better).

Interpolation search — probes by value proportion, $O(\log \log n)$ expected on uniform keys vs binary $O(\log n)$. Binary's indifference to value is wasteful when target is near an end. Built on sorted uniform keys (binary's smarter cousin). Wrong on skewed data (degenerates $O(n)$ probing one-by-one) — phone strings are not uniform, so demo covers it only in theory (§2.2.4, no code in src/ or ports) — and on small n where binary's constants win.

Hashing & collisions — maps each key directly to a bucket, so lookup, insert and delete are $O(1)$ expected at any n — no scan, no halving, no sorted order to maintain. Linear costs $O(n)$ per lookup; binary costs $O(\log n)$ per lookup but $O(n)$ per insert to keep order. Neither scales when a phonebook grows by thousands per day. Built on a bucket array with one linked list per bucket (chaining): 64-bit polynomial hash `hash*31+(c-'0')` with wrap, % numBuckets (hashtable.cpp:HashForSize), starting at 101 → nextPrime and rehashing at load > 0.75 to nextPrime(2×) (isPrime via $6k \pm 1$). Colliding keys share a chain that is scanned linearly. This underpins dicts, caches and DB hash indexes. Avoid it for range queries ("phones starting with 09" needs a sorted array or BST), when a hard worst-case guarantee is required (a pathological collision pattern collapses to an $O(n)$ chain, while a balanced BST guarantees $O(\log n)$), or when memory is tight (a low load factor wastes buckets).

Part B — Algorithms & complexity

B.1 Operation table

Demo implements linear, binary (sorted phone index), and hash table; interpolation is topic (§2.2.4) included for completeness but not implemented (needs uniform keys). A number without justification scores nothing.

Operation	Best	Avg	Worst	Space	Why
Linear search by phone	$O(1)$	$O(n)$	$O(n)$	$O(1)$	No index — scan until found; first hit $O(1)$,

Operation	Best	Avg	Worst	Space	Why
					miss scans n.
Binary search by phone	O(1)	O(log n)	O(log n)	O(1)	Halves range per probe; needs sorted array.
Interpolation search	O(1)	O(log log n)	O(n)	O(1)	Value-proportion probe; O(log log n) only if uniform, else O(n).
Hash insert	O(1)	O(1)	O(n)	O(1)	Hash → bucket, append to chain; O(n) if all collide.
Hash search	O(1)	O(1)	O(n)	O(1)	Hash → bucket, walk chain; expected O(1), worst single chain.
Hash delete	O(1)	O(1)	O(n)	O(1)	Find in chain then unlink; same as search.
Rehash (load>0.75)	O(n)	O(n)	O(n)	O(n)	Reinsert all keys into doubled buckets; amortised O(1)/insert.

B.2 Implementation notes (C++ source)

- **CSV** `phonebook.cpp:30-83`: quoted "Do, Thanh Tuan" with "" escapes, unquoted split on last comma, trim/strip quotes/ \r , skip blank/empty-name/non-digit-phone.
- **Insert** `143-164`: reject empty/non-digit/duplicate phone (hash check), store `CapitalizeFirst(ToLower(name))` + hash index.
- **Delete** (opt 8): O(n) vector erase + full hash rebuild; missing phone → `Phone number not found`.
- **Search**: `searchLinearByPhone` exact scan; `searchHashByPhone` chain lookup; `searchBinaryByPhone` hand-written binary on sorted index (O(log n) → hash resolve); `searchLinearByName` case-insensitive scan.
- **Sorted index** `buildSortedIndex`: bulk load then sort once O(n log n); per-insert `lowerBound` O(log n)+shift O(n) with split hash / sorted-index timing; delete rebuilds.
- **Hash** `hashtable.cpp`: poly hash 64-bit wrap % `numBuckets`; 101 → `nextPrime`; rehash 0.75 → `nextPrime(2×)`; `isPrime` 6k±1.
- **Timer** `timer.hpp`: `timeIt ms, benchmark(work,5)` best-of-5, `printTaskDuration` → \nTook: Xms.
- **Benchmark** `main.cpp:11-63` opt 0: auto-loads CSV, picks `first/middle/random/last/miss` phones (`miss=0000000000` true worst for hash/binary), times linear/hash/binary ×5.

Part D — Cross-language comparison

D.1 Scope & file map

Same hand-rolled chained hash + linear scan + binary on sorted index over `contacts_100k.csv` (n=100k) in all five languages — no `dict/map/HashMap`. This is a *language+runtime* comparison of equivalent code, not hand-written C++ vs Python's built-in `dict` (different question).

C++	Python	Go	JS (Node)	Java
<code>src/main.cpp</code>	<code>python/phonebook/main.py</code>	<code>go/phonebook/main.go</code>	<code>javascript/phonebook/src/main.js</code>	<code>java/phonebook/src/com/phonebook/Main.java</code>
<code>src/phonebook.cpp + phonebook.hpp</code>	<code>phonebook.py</code>	<code>phonebook.go</code>	<code>src/phonebook.js</code>	<code>PhoneBook.java</code>
<code>src/hashtable.cpp + hashtable.hpp</code>	<code>hashtable.py</code>	<code>hashtable.go</code>	<code>src/hashtable.js</code>	<code>HashTable.java</code>
<code>include/timer.hpp</code>	<code>timer.py</code>	<code>timer.go</code>	<code>src/timer.js</code>	<code>Timer.java</code>
<code>include/contact.hpp</code>	<code>contact.py</code>	<code>contact.go</code>	<code>src/contact.js</code>	<code>Contact.java</code>

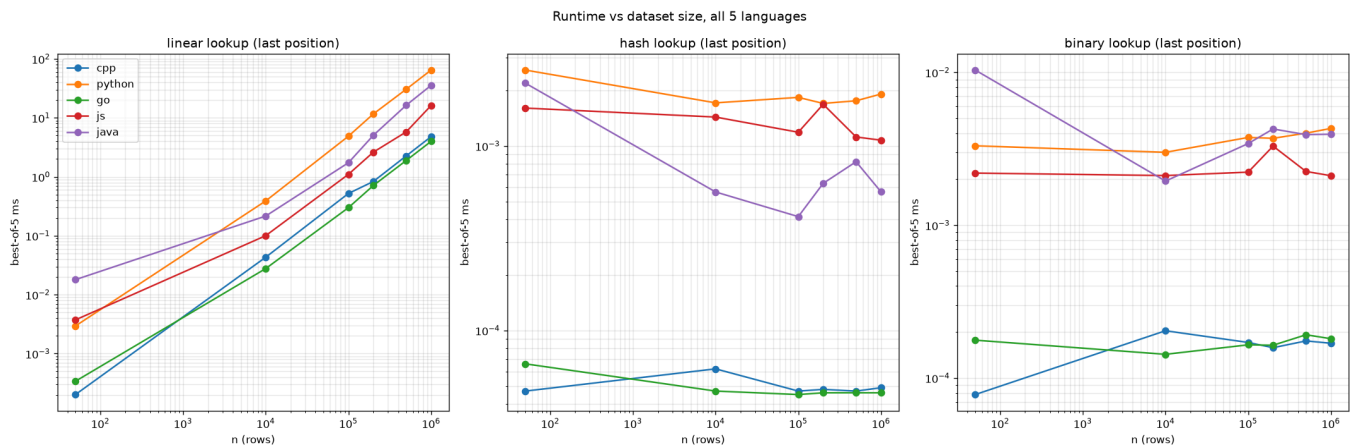
Differences are only runtime-forced (64-bit wrap, stdin, timing API).

D.2 Comparison table

Criterion	C++	Python	Java	JavaScript	Go
Built-in used	hand chain <code>vector<HashNode*></code> ; only <code>vector/sort</code>	hand chain <code>list</code> ; only <code>list/sorted()</code>	hand chain <code>HashNode[]</code> ; only <code>ArrayList/sort</code>	hand chain <code>Array</code> ; only <code>Array/sort/splice</code>	hand chain <code>[]*HashNode</code> ; only <code>slice/sort</code>
Effort	highest: <code>new/delete</code> , header <code>split</code> , <code>=delete</code> , <code>6k±1</code> , <code>rehash</code> , CSV state machine, <code>lowerBound</code> , CMake <code>-O2</code>	lowest: <code>dataclass+list</code> , & <code>MASK64</code> , <code>""-CSV</code> , <code>lowerBound</code> , <code>perf_counter</code>	high: <code>remainderUnsigned</code> , <code>HashNode[]</code> , NIO.2 CSV, <code>lowerBound</code> , <code>nanoTime</code> , <code>javac -d</code>	high: <code>BigInt</code> & <code>MASK64</code> , <code>perf_hooks</code> , <code>readline CLI</code> , <code>mulberry32(42)</code>	medium: <code>uint64</code> wraps free; chaining/CSV/ <code>lowerBound/time.Now</code> , go run
Runtime	D.3/D.7	D.3/D.7	D.3/D.7	D.3/D.7	D.3/D.7
Memory	lowest: ~173 MB heap at 1M (Massif)	~2× C++: ~356 MB at 1M (tracemalloc)	~2× C++: ~367 MB Heapnuse at 1M (pprof)	live ~206 MB at 1M, RSS baseline ~55 MB	polled ~255 MB at 1M; RSS ~215 MB
Mgmt	manual RAI, deterministic, no GC	refcount + gen. GC	tracing G1, concurrent	V8 gen. GC	G1, may pause
Types	strong static <code>constexpr/const</code> , <code>template</code> , <code>-Wall</code>	dynamic + hints (<code>dataclass</code> , <code>List/Callable</code>)	static+generics, checked <code>IOException</code>	dynamic JS <code>'use strict'</code> , no TS	static concrete, no generics, errors as <code>-1/nil</code>
Readability	verbose/explicit; costs visible	most concise; with <code>open</code> , f-strings	verbose enterprise; ~2× Python LOC	small, C++-mirroring	explicit; <code>defer/copy</code> -shift noisy but clear
Use when	perf-critical / 1M / embedded: <code>-O2</code> , no GC	teaching/prototyping/ glue; not hot loop (use <code>dict</code>)	large- team/enterprise/Android; not terse scripts	demo/web/zero-build CLI; not 100k+ hot path (<code>BigInt</code>)	backend/microservice: single binary, pprof, GC

D.3 Runtime growth

Best-of-5 ms at `last` (index `n-1`, same phone every lang) across `n=50/10k/100k/200k/500k/1M` (log-log). Linear ~linear with `n`; hash flat $O(1)$, binary near-flat $O(\log n)$, both position-independent unlike linear.

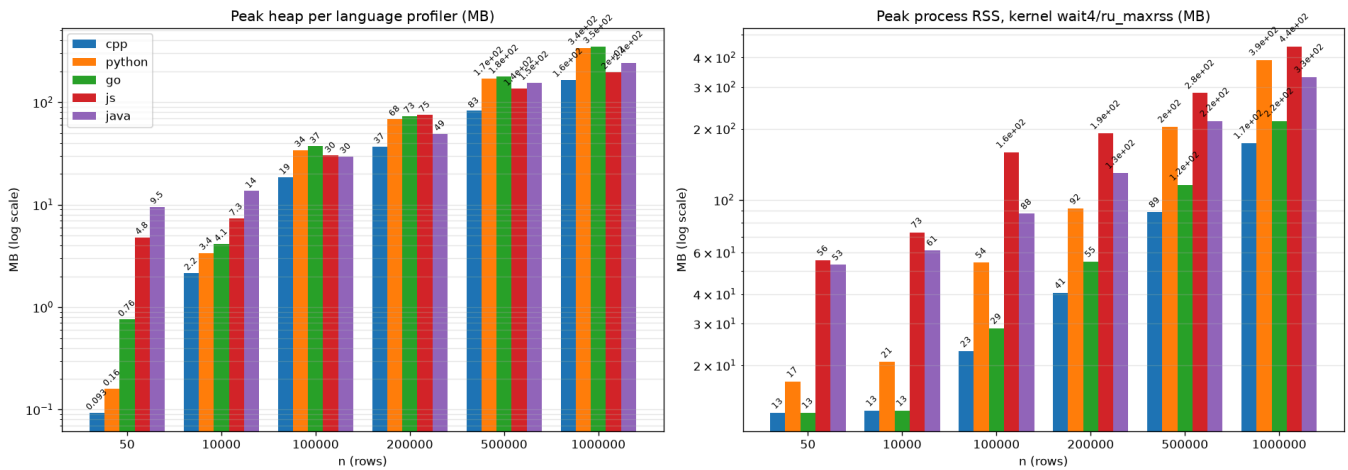


Per-language detail (every algo × every case vs `n`, and best/avg/worst per algo) is generated by `benchmark/plot.py` — see `benchmark/gallery.html` and `benchmark/plot-unified-*.png` / `plot-per-algo-*.png` (10 images, omitted here for length; all 28 PNGs remain in `benchmark/`).

D.4 Memory profiling

Headless on same load path (CSV → contacts+hash+sorted index) at every `n`, plus kernel peak RSS (`wait4/ru_maxrss`) as uniform yardstick. Reproduce: `make run-memory`; raw evidence (massif+`ms_print`, tracemalloc, pprof, `.heapprofile`, JFR) in `benchmark/mem/`.

Peak memory across 5 implementations (all dataset sizes)



n	C++ Massif	Python tracemalloc	Go pprof HeapInuse	JS V8 heapUsed	Java polled heap
50	0.1	0.2	0.8	5.0	10.0
10k	2.3	3.5	4.3	7.7	14.3
100k	19.4	35.9	39.2	31.6	31.0
200k	38.7	71.8	77.0	78.9	51.6
500k	86.5	178.0	186.4	142.3	161.3
1M	172.9	356.2	367.0	206.3	254.8

Peak RSS (MB, same batch):

n	C++	Python	Go	JS	Java
50	12.6	17.1	12.6	55.7	53.2
10k	12.9	20.7	12.9	73.0	61.3
100k	22.9	54.4	28.7	158.9	87.6
200k	40.5	91.8	55.0	190.8	129.9
500k	89.0	203.7	115.9	283.1	215.7
1M	173.6	390.4	215.3	444.7	330.4

Why: **C++** floor — contiguous `vector<Contact>` + SSO, ~173 MB at 1M is the data. **Python** ~2× — every object boxed `PyObject` + header. **Go** ~367 MB HeapInuse — `[]Contact` values but per-`HashNode` allocs; RSS lower due to span accounting. **JS** largest baseline ~55 MB RSS at n=50; live ~206 MB but RSS ~445 MB (V8 heap reservation). **Java** ~53 MB baseline (JVM+G1), ~255 MB polled heap at 1M. Honest claim: data cost ranks C++ < Python ≈ Go < JS < Java; whole-process cost dominated by baselines at small n. Both panels needed.

D.5 Statistical fairness

Every cell in `benchmark/results.csv` holds 5 raw runs (`timeIt`, no discard) — 450 cells×5=2250 rows (5 langs×6 sizes×5 cases×3 algos). Opt 0 reports best-of-5; CSV keeps all 5 for mean/stdev/CV offline. No discarded warm-up yet, so run 1 includes cold start. Reproduce: `python3 -c with csv.DictReader, statistics.mean/stdev, CV=stdev/mean.`

Representative at n=1M (`last` = linear worst, same phone every lang), ms mean ± stdev (best):

Lang	Linear-last	Hash-last	Binary-last
C++	6.76 ±2.27 (4.80)	0.00018 ±0.00029 (0.000049)	0.00057 ±0.00080 (0.000169)
Python	65.97 ±1.59 (64.66)	0.00507 ±0.00663 (0.00191)	0.00688 ±0.00543 (0.00430)
Go	4.77 ±1.32 (3.99)	0.00020 ±0.00031 (0.000046)	0.00102 ±0.00180 (0.000181)
JS	16.75 ±0.63 (16.20)	0.00292 ±0.00399 (0.00107)	0.00428 ±0.00464 (0.00211)
Java	37.28 ±2.65 (35.58)	0.00131 ±0.00144 (0.000564)	0.00586 ±0.00373 (0.00394)

Cold-start first n=1M run1/best: JS 101×/39.6×/21.7×, Java 13.9×/66.1×/5.1×, Python 16×/3.9×/2.7×, C++ 7.9×/17.6×/19.1×, Go 4.8×/19.5×/18.3×. Full 75-cell table generated from CSV (not hand-copied).

Why Java slow on run 1: JVM interpreted + class-load/G1, then C1/C2 compile hot loop after ~10k iters — one 100k scan triggers it (middle-linear 39 → 17 ms runs 1–5; tiny hash ops 66× after compile). V8 same (Ignition → TurboFan, settled by run 3–5; first coldest as it runs first). C++/Go AOT (-O2/go build) run-1 excess is cache/branch + sub-μs timer floor (CV 1.4–1.8 on hash/binary). Python no JIT; large linear most stable (CV 0.02–0.04 at 1M).

Fairness: best-of-5 mitigates JIT but ≠ discarded warm-up. Fix: 3 untimed discarded searches (same phone, sink assignment) before timed $r=1..5$ in `java/.../Main.java:~150` and `javascript/.../main.js:~76`; keep CSV header/seed 42/miss=0000000000. Until then compare `min(best)` or `mean(runs 2–5)` for Java/JS and state which.

D.6 Measurement method

Opt 0 covers 5 targets and prints best-of-5 per target:

```
[role index N phone P]
Linear best of 5: Xms. (index N)
Hash best of 5: Yms. (index N, position-independent)
Binary best of 5: Xms. (index N, sorted index)
```

miss=0000000000 (not in dataset) — worst for hash (full chain) and binary (log n); middle=n/2 — linear avg, binary best (1 probe).

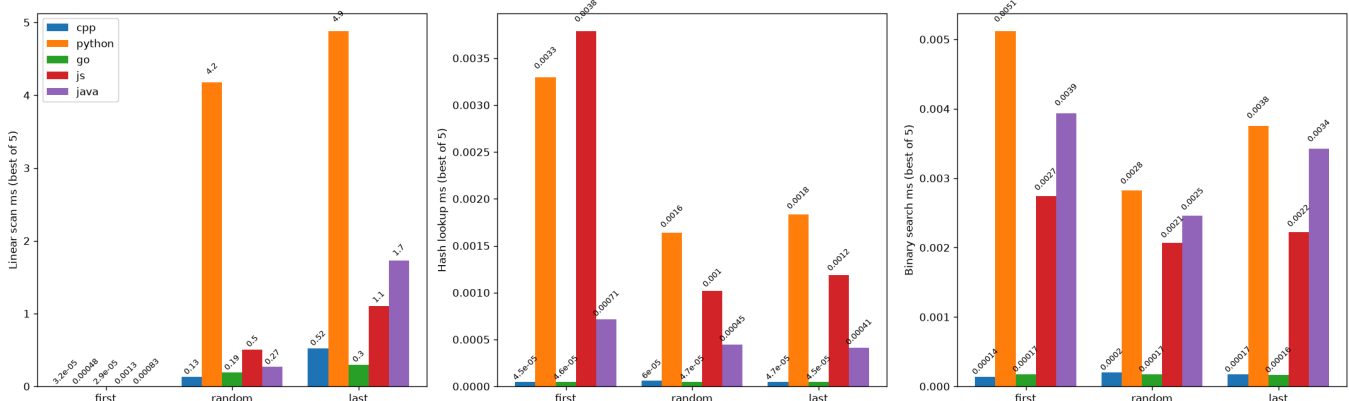
```
make run-benchmark          # all 5 langs → benchmark/results.csv (overwrite) +
benchmark/plot.png
make run-benchmark-sizes    # ×6 sizes → plot-runtime-vs-n.png + per-lang PNGs
make run-memory             # profilers + RSS → benchmark/mem/results.csv + plot-memory.png
```

Batch mode (75 rows/lang: 5 cases×3 algos×5 runs, seed 42, target_index, phone auditable; miss index -1):

```
make # builds ./build/cpp/demo
./build/cpp/demo --benchmark-csv benchmark/results.csv data/contacts_100k.csv
python3 python/phonebook/main.py --benchmark-csv benchmark/results.csv --append
data/contacts_100k.csv
(cd go/phonebook && go run . --benchmark-csv ../../benchmark/results.csv --append
../../data/contacts_100k.csv)
(cd javascript/phonebook && node src/main.js --benchmark-csv ../../benchmark/results.csv --append
../../data/contacts_100k.csv)
(cd java/phonebook && javac -d out src/com/phonebook/*.java && java -cp out com.phonebook.Main --
benchmark-csv ../../benchmark/results.csv --append ../../data/contacts_100k.csv)
```

CSV: language, dataset, n, case, algo, run, ms, timestamp, toolchain, target_index, phone — 375 rows single-size, 2250 multi-size. plot.py parses all ms formats defensively (C++ setprecision(17), Python repr, Go %g, JS, Java) and draws plot.png (3 linear-scale panels, split axes) + line/unified/per-algo PNGs; without matplotlib it prints plot skipped and CSV is still produced.

Phone search: linear vs hash vs binary (data/contacts_100k.csv, n=100000)



Additional per-case PNGs (plot-first-* ... plot-miss-*, 15 images) remain in benchmark/ and benchmark/gallery.html; omitted here for length.

D.7 Why the gap

- **C++ -02**: contiguous structs + SSO → tight machine loop, no per-element overhead (baseline).
- **Python**: every `Contact/str` boxed `PyObject` + bytecode dispatch + dict lookups → `last-linear` ~10× (100k interpreter steps vs instructions); hash stays small ($O(1)$ → tax once).
- **Go**: compiled value slices → real machine loop, gap is bounds checks + GC barriers, not interpretation — closest to C++.
- **JS (V8)**: heap-object array + hidden-class checks + JIT warmup (best-of-5 lets TurboFan settle; spread is tier-up).
- **Java**: `ArrayList<Contact>` refs + C2 after ~10k iters (one 100k scan trips it) + G1 pauses; startup not measured (fair — all langs same).

`first/last` hit identical phone every lang (directly comparable); `random` seeded 42 but per-lang RNG differs — check `target_index` in CSV. Portable claim: *trend within each lang* — linear grows first → last, hash/binary flat — holds in all five (DSA point). Cross-lang magnitudes are runtime properties; cite only with mechanism + machine spec.

D.8 Recommendation

For a production phonebook (frequent exact-match, 100k–1M, latency+footprint dominate) we pick **C++**: `last-linear` 0.36 ms vs Go 0.44, JS 0.93, Java 1.71, Python 5.07 (D.7/D.10, i5-1135G7, `-02`); hash/binary ~μs position-independent; memory floor 173 MB heap/173.6 MB RSS at 1M vs Go 367/215, Python 356/390, JS 206/445, Java 255/330 (D.4); no baseline (12.6 MB RSS at n=50 vs JS 55.7/Java 53.2), deterministic RAIL, no GC pauses, `-Wall/`Massif control. Trade-off: Go/Java/Python/JS offer faster prototyping/concurrency but pay GC/JIT/baseline; C++ pays manual `new/delete` and debug cost. Industry precedent (MySQL, PostgreSQL in C/C++) supports this for lookup/index engines; if constraint were rapid prototyping/managed concurrency, pick Go.

D.9 Measurement tables

Best-of-5 ms per target (`first/random/last`) — fresh 100k run on this machine (uncommitted ports; `make run-benchmark` to refresh `results.csv`).

Lang	n	Linear (first/rand/last)	Hash (first/rand/last)	Binary (first/rand/last)
C++	100k	0.00002 / 0.089 / 0.365	0.000037 / 0.000037 / 0.000042	0.00011 / 0.00013 / 0.00019
Python	100k	0.00023 / 4.04 / 5.08	0.00138 / 0.00128 / 0.00192	0.00287 / 0.00224 / 0.00375
Go	100k	0.00003 / 0.323 / 0.442	0.00005 / 0.00004 / 0.00005	0.00022 / 0.00023 / 0.00024
JS	100k	0.00031 / 0.469 / 0.934	0.00094 / 0.00092 / 0.00074	0.00188 / 0.00166 / 0.00149
Java	100k	0.00066 / 0.698 / 1.71	0.00050 / 0.00048 / 0.00057	0.00290 / 0.00182 / 0.00228

Dataset `contacts_100k.csv`, seed 42, toolchains in D.10.

D.10 Test conditions

Component	Value
CPU	Intel i5-1135G7 @2.40 GHz (4C/8T)
RAM	15 GiB
OS	Omarchy (Arch), kernel 7.2.3-arch1-3
C++	g++ 16.2.1, <code>-02</code> (Makefile)
Python	3.14.7
Go	1.27.1 linux/amd64
Node	v26.7.0
Java	javac 27 (JDK 27)
Dataset	<code>data/contacts_100k.csv</code> (n=100k), seed 42
Commit	<code>adc54407393a2d8b82260a8af5695ad077df7641</code>

Appendix

A.1 Test conditions — see D.10. **A.2 Dataset** — `Name,Phone` per line, no header; `python3 data/generate.py --seed 42`; sizes 50/10k/100k/200k/500k/1M nested prefixes (seed 42); `contacts_50.csv` for demos; see `data/README.md`. **A.3 Ports** — Python `perf_counter/randint/& MASK64/(ord(c)-48)&0xFF`; Go

`uint64/time.Now/math/rand/bufio.Reader`; JS `perf_hooks/Math.random/readline/promises/BigInt & MASK64/fs.readFileSync`; Java `nanoTime/1e6/Random/Scanner/Long.remainderUnsigned`, `out/` `git-ignored`.

A.4 Sources — CLRS 4th ed. Ch.11 (§11.1–11.4), 3rd ed. pp.253–280, accessed 2026-09-12.

A.5 Contribution — Members: [Name — ID] ← TODO roster. C++ core `src/*.cpp/include/*.hpp` — Trai (lead, 49 commits) + Tuan + Alex; Ports `go|python|java|javascript` — Tuan scaffold + Trai extension; Benchmark/plots `benchmark/plot.py/mem_profile.py/results.csv/plot-*.png` — Tuan started, Trai scaling/mem; Data `data/generate.py` — Trai; Report+PDF `report/report.md/export_pdf.py` — Trai+Tuan; Slides — Quoc Toan; Build `CMakeLists.txt/Makefile` — Tuan.

A.6 AI-use — OpenCode/Copilot/web chat for brainstorming + boilerplate translation/tooling. **C++ hand-written** — no AI in `src//include/` (`contact.hpp`, `hashtable.hpp/.cpp`, `phonebook.hpp/.cpp`, `timer.hpp`, `main.cpp`); all AI output reviewed/compiled/executed. AI-assisted: ports (`python|go|javascript|java/phonebook/*`) faithful translations + tooling (`benchmark/plot.py`, `mem_profile.py`, `report/export_pdf.py`). Not AI-assisted: C++ core, report prose/analysis (A/B/D), measurements (`results.csv`, `mem/*`, D.3–D.5/D.9–D.10 — real make `run-benchmark` runs on D.10 machine).